Seasonal time series

2026 April 06

Three examples of seasonal time series: Vancouver monthly average temperature, Vancouver total monthly precipitation, log airpassengers.

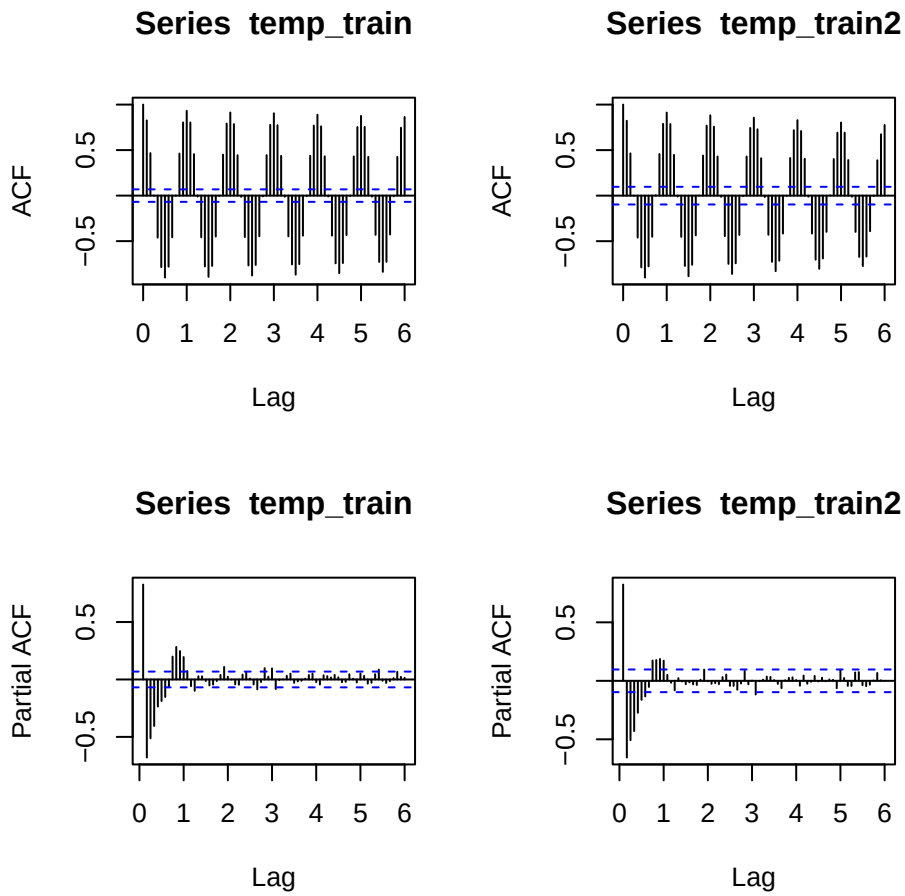
This Rmd file reviews many things; also there are comparisons for models that should be good for short-term forecasting and models that might generate data similar to the given data series. Also, there are diagnostics for the estimated innovations.

Input data set Vancouver temperature and precipitation data 1938 to 2023

```
v_precip_temp = read.csv("vanc-prec-temp.csv", header=T)

temp = ts(v_precip_temp$meantemp, start=c(1938,1), frequency=12)
temp_train = window(temp, end=c(2005,12)) # 68 years
temp_train2 = window(temp, end=c(1971,12)) # 34 years
temp_holdout = window(temp, start=c(2006,1))
ntrain = length(temp_train)

precip = ts(v_precip_temp$totprecip, start=c(1938,1), frequency=12)
precip_train = window(precip, end=c(2005,12)) # 68 years
precip_train2 = window(precip, end=c(1971,12)) # 34 years
precip_holdout = window(precip, start=c(2006,1))
```

Monthly average temperature

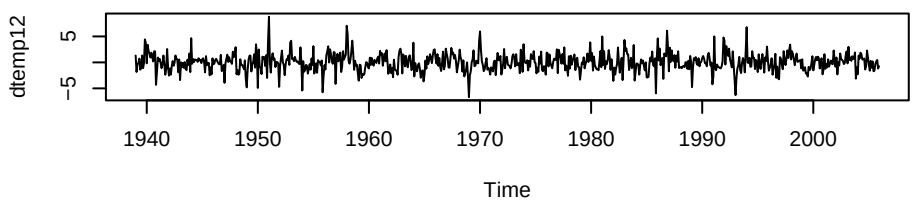
```
par(mfrow=c(2,2))
temp_acf = acf(temp_train, lag.max=72)      # length ntrain
temp_acf1 = acf(temp_train2, lag.max=72)    # length ntrain/2
temp_pacf = pacf(temp_train, lag.max=72)    # length ntrain
temp_pacf2 = pacf(temp_train2, lag.max=72)  # length ntrain/2
```

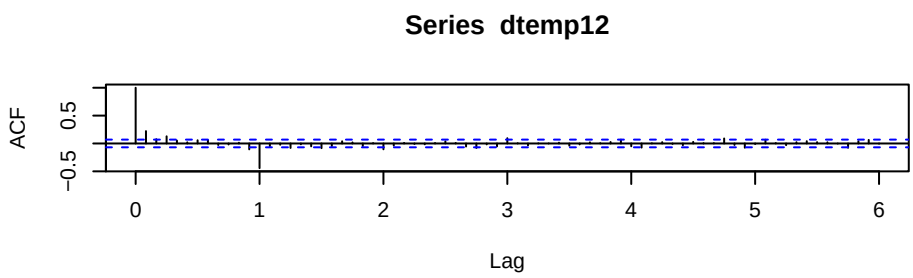


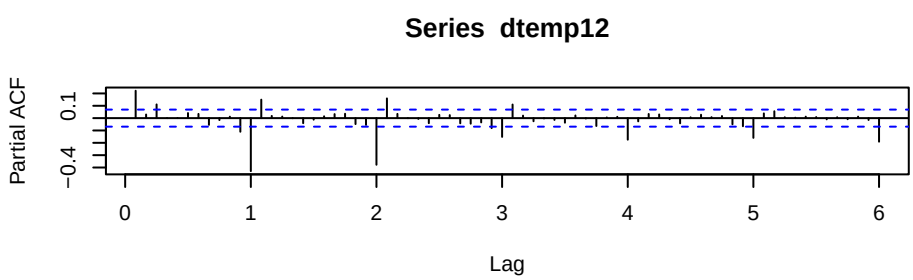

```
# Dashed lines on acf and pacf plots are at +/- 2/sqrt(n),
# where n = length of time series, see theory document for explanation.

# What can be concluded from the 4 plots?

temp_acfv = c(temp_acf$acf)
lags = 1+seq(6,72,6)  # every 6th serial correlation, first is 1 for lag 0

# acf values for average temperature at lags of 6 months apart
print(round(temp_acfv[lags],3))
#>  [1] -0.900  0.932 -0.892  0.913 -0.878  0.905 -0.868  0.888 -0.852  0.876
#> [11] -0.838  0.863

# Decrease in absolute value is very slow; Box-Jenkins ARIMA suggest seasonal differencing

acf12 = temp_acfv[13]  # lag12 serial correlation
print(round(acf12^(1:6),3))
#> [1] 0.932 0.868 0.809 0.753 0.702 0.654

# Seasonal acf is decreasing at a slower rate than AR(1) with rho1=0.932

# Winters additive seasonal exponential smoothing
aseas_temp = HoltWinters(temp_train, seasonal="additive")
```

```r
print(aseas_temp)
#> Holt-Winters exponential smoothing with trend and additive seasonal component.
#>
#> Call:
#> HoltWinters(x = temp_train, seasonal = "additive")
#>
#> Smoothing parameters:
#>  alpha: 0.1044861
#>  beta : 0
#>  gamma: 0.1150224
#>
#> Coefficients:
#>             [,1]
#> a   10.857629893
#> b    0.001056235
#> s1  -6.363302431
#> s2  -5.639900083
#> s3  -3.531740405
#> s4  -0.882535606
#> s5   2.403918094
#> s6   5.312445001
#> s7   7.552694490
#> s8   7.655801063
#> s9   4.426033148
#> s10 -0.041641446
#> s11 -4.125814513
#> s12 -6.340237418

# If Winters additive seasonal exponential smoothing forecasts well, and
# beta is in (0,1), the corresponding ARIMA model has differencing at lag 1 and 12
# For beta=0 and B_t=0 for all t: it can be shown that seasonal exponential
# smoothing leads to an ARIMA model with only differencing at lag 12

# Diagnostics for differencing at lag 12
dtemp12 = diff(temp_train, lag=12)

par(mfrow=c(3,1))
plot(dtemp12)
print(window(dtemp12,start=c(1939,1),end=c(1948,12)))
#>        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
#> 1939  1.31 -1.84 -0.32 -0.13  0.73 -1.40 -0.98  1.32 -1.11 -0.17  4.43  3.67
#> 1940 -0.18  3.38  2.19  1.69  0.94  1.87  0.96  0.19  2.14  2.10 -4.32 -1.85
#> 1941  0.94  0.27  0.66 -0.09 -1.12 -0.29  1.33 -0.39 -2.37 -1.24  2.58 -0.86
#> 1942 -2.55 -1.17 -2.37 -0.70  0.25 -0.64 -0.10  1.33  0.67  0.61 -1.89  0.56
#> 1943 -3.41  0.01 -0.92 -0.09 -1.04  0.07 -1.29 -1.49  0.53 -0.50  1.70 -1.86
#> 1944  4.67 -0.26  0.66  0.07  0.57  0.50  0.46  0.11  0.81  0.38  0.46 -0.13
#> 1945  0.66  0.79 -0.01 -1.92  1.21 -0.54 -0.26 -0.09 -2.49 -0.94 -2.31  0.78
#> 1946 -0.37 -0.21  0.49  0.69  0.29 -0.07 -0.35 -0.32  1.45 -1.67 -1.30 -0.50
#> 1947 -3.93 -0.19  0.68  1.48  0.09  1.00  0.20  0.21 -0.28  2.06  1.01  1.27
#> 1948  2.93 -2.28 -1.46 -2.38 -2.11  0.61 -0.43 -0.49 -0.86 -1.32  0.39 -3.26

# Does this series appear stationary?
```

```r
# What are reasons for the original series to be considered non-stationary?

dtemp12_acf = acf(dtemp12, lag.max=72)
dtemp12_pacf = pacf(dtemp12, lag.max=72)
```

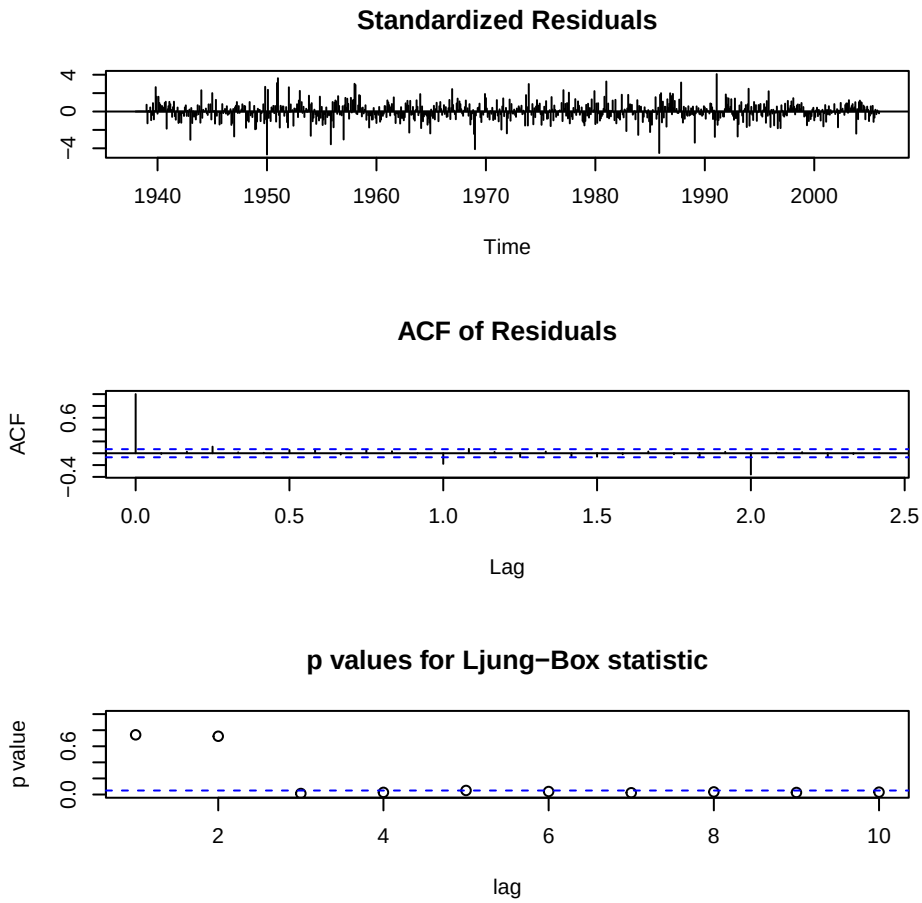




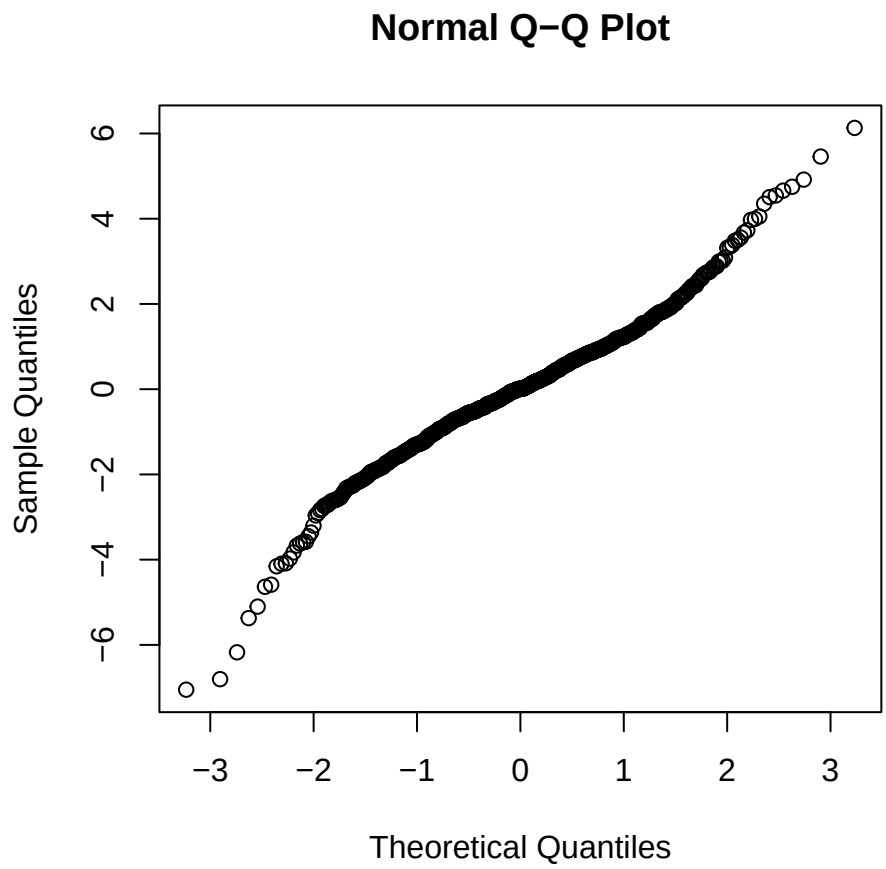



```r
dtemp12_acfv = c(dtemp12_acf$acf)
dtemp12_pacfv = c(dtemp12_pacf$acf)

# acf values for seasonal difference of temperature at lags of 6 months apart
print(round(dtemp12_acfv[lags],3))
#>  [1]  0.058 -0.443 -0.085 -0.105  0.038  0.095 -0.040 -0.051  0.029 -0.011
#> [11]  0.024 -0.004

print(round(dtemp12_pacfv[lags-1],3))
#>  [1]  0.041 -0.429 -0.013 -0.378  0.027 -0.152 -0.037 -0.175  0.007 -0.160
#> [11]  0.009 -0.191


# Compare fits of stationary and nonstationary models
library(forecast)
auto_temp_stationary = auto.arima(temp_train, seasonal=T,stationary=T)

auto_temp_nonstationary = auto.arima(temp_train, seasonal=T,stationary=F)
```

```r
print(auto_temp_stationary)
#> Series: temp_train
#> ARIMA(5,0,1)(0,0,2)[12] with non-zero mean
#>
#> Coefficients:
#>          ar1      ar2      ar3      ar4      ar5      ma1    sma1    sma2
#>       1.0178  -0.2241  -0.0686  -0.1849  -0.1309  -0.4324  0.2080  0.0823
#> s.e.  0.0634   0.0622   0.0504   0.0495   0.0451   0.0541  0.0384  0.0339
#>          mean
#>       10.0302
#> s.e.   0.0668
#>
#> sigma^2 = 2.403:  log likelihood = -1513.77
#> AIC=3047.54   AICc=3047.82   BIC=3094.59

print(auto_temp_nonstationary)
#> Series: temp_train
#> ARIMA(1,0,0)(1,1,0)[12] with drift
#>
#> Coefficients:
#>          ar1     sar1   drift
#>       0.2448  -0.4571  0.0016
#> s.e.  0.0342   0.0314  0.0040
#>
#> sigma^2 = 2.272:  log likelihood = -1470.69
#> AIC=2949.37   AICc=2949.42   BIC=2968.13

# Unlike lm and glm output, no P-values are given
# A rough "assessment" or "diagnostic" for non-significance is whether
# |coefficient/SE| < 2 =~ 1.96.
# If the model is correct, then the sample distribution of each
# estimated coefficient as a random variable is asymptotically normal.
# But be aware of the problems with hypothesis testing for large sample sizes
# or if the model is chosen based on a rule like forward selection or min AIC.

# drift term is not significant?
# Thw model with drift implies that E[diff(Y_t,12)] = mean_drift

# Note that the fitted model with seasonal differencing has smaller AIC

# tsdiag function has diagnostics to check if estimated innovations
# behave close to white noise
tsdiag(auto_temp_nonstationary)

# See theory document for definition of Ljung-Box statistic

tsdiag(auto_temp_nonstationary)
```




```
par(mfrow=c(1,1))
qqnorm(auto_temp_nonstationary$residuals)
```



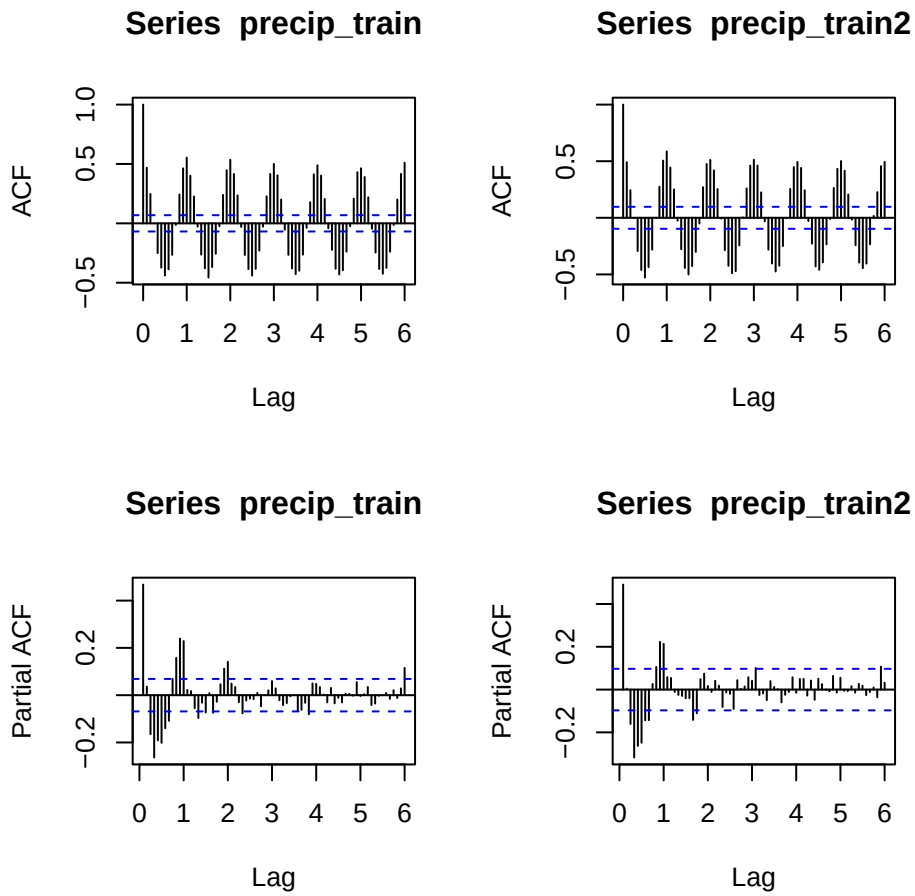

```
# QQ plot for normality, maybe slightly heavier-tailed than Gaussian

# Normality and homoscedasticity of the innovations are assumptions
# for valid forecast intervals

fc = forecast(auto_temp_nonstationary)
options(digits=3)
print(fc)
#>          Point Forecast Lo 80 Hi 80  Lo 95 Hi 95
#> Jan 2006           3.82  1.89  5.76  0.869  6.78
#> Feb 2006           5.04  3.05  7.03  1.997  8.08
#> Mar 2006           8.25  6.26 10.24  5.203 11.30
#> Apr 2006          10.61  8.62 12.60  7.565 13.66
#> May 2006          14.28 12.28 16.27 11.229 17.32
#> Jun 2006          16.40 14.41 18.40 13.357 19.45
#> Jul 2006          18.89 16.90 20.88 15.845 21.94
#> Aug 2006          19.16 17.17 21.15 16.115 22.21
#> Sep 2006          14.61 12.61 16.60 11.559 17.65
#> Oct 2006          11.11  9.11 13.10  8.060 14.15
#> Nov 2006           6.22  4.22  8.21  3.170  9.26
#> Dec 2006           4.97  2.98  6.96  1.923  8.02
#> Jan 2007           3.80  1.55  6.05  0.356  7.24
#> Feb 2007           4.73  2.46  6.99  1.262  8.19
```

```
#> Mar 2007           8.33  6.06 10.59  4.860 11.79
#> Apr 2007          10.41  8.15 12.68  6.947 13.88
#> May 2007          14.33 12.07 16.60 10.865 17.80
#> Jun 2007          16.06 13.80 18.33 12.597 19.53
#> Jul 2007          18.57 16.30 20.83 15.099 22.03
#> Aug 2007          19.10 16.83 21.37 15.634 22.57
#> Sep 2007          14.66 12.40 16.93 11.195 18.13
#> Oct 2007          11.22  8.95 13.49  7.751 14.69
#> Nov 2007           5.98  3.72  8.25  2.518  9.45
#> Dec 2007           4.83  2.56  7.10  1.361  8.30

options(digits=7)

# Compare predict function after arima()
arima_temp = arima(temp_train,order=c(1,0,0),seasonal=list(order=c(1,1,0),period=12))
print(arima_temp)
#>
#> Call:
#> arima(x = temp_train, order = c(1, 0, 0), seasonal = list(order = c(1, 1, 0),
#>     period = 12))
#>
#> Coefficients:
#>          ar1     sar1
#>       0.2450  -0.4570
#> s.e.  0.0342   0.0314
#>
#> sigma^2 estimated as 2.264:  log likelihood = -1470.76,  aic = 2947.52

(p = predict(arima_temp,n.ahead=5))
#> $pred
#>            Jan       Feb       Mar       Apr       May
#> 2006  3.802810  5.012486  8.222895 10.585002 14.248328
#>
#> $se
#>           Jan      Feb      Mar      Apr      May
#> 2006 1.504681 1.549189 1.551821 1.551979 1.551988

cv80 = qnorm(0.90); cv95 = qnorm(0.975)
pred_table = cbind(p$pred, p$pred-cv80*p$se, p$pred+cv80*p$se,
  p$pred-cv95*p$se, p$pred+cv95*p$se)
colnames(pred_table)= c("pointfc","Lo80","Hi80","Lo95","Hi95")
round(pred_table,2)
#>          pointfc  Lo80  Hi80  Lo95  Hi95
#> Jan 2006    3.80  1.87  5.73  0.85  6.75
#> Feb 2006    5.01  3.03  7.00  1.98  8.05
#> Mar 2006    8.22  6.23 10.21  5.18 11.26
#> Apr 2006   10.59  8.60 12.57  7.54 13.63
#> May 2006   14.25 12.26 16.24 11.21 17.29

# The drift parameter from auto,arima has a very small effect.

length(auto_temp_nonstationary$residuals)
#> [1] 816
```

```r
# Estimated innovations have weak serial correlations
round(c(acf(auto_temp_nonstationary$residuals,plot=F,lag.max=12)$acf),3)
#>  [1]  1.000 -0.011  0.026  0.111  0.015  0.007  0.053  0.060 -0.021  0.052
#> [11]  0.034 -0.002 -0.180

Box.test(auto_temp_nonstationary$residuals)
#>
#>  Box-Pierce test
#>
#> data:  auto_temp_nonstationary$residuals
#> X-squared = 0.10754, df = 1, p-value = 0.743

Box.test(auto_temp_nonstationary$residuals,lag=3, type="Ljung-Box")
#>
#>  Box-Ljung test
#>
#> data:  auto_temp_nonstationary$residuals
#> X-squared = 10.817, df = 3, p-value = 0.01276
```

Monthly total precipitation

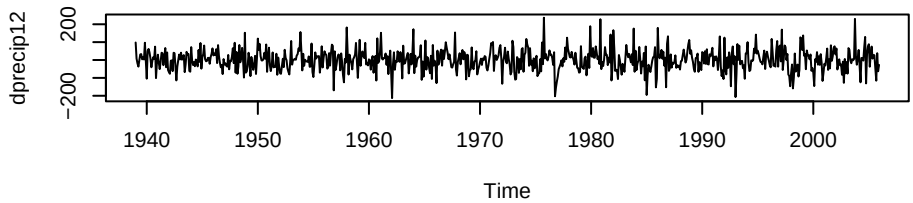
```r
par(mfrow=c(2,2))
precip_acf = acf(precip_train, lag.max=72)
precip_acf2 = acf(precip_train2, lag.max=72)
precip_pacf = pacf(precip_train, lag.max=72)
precip_pacf2 = pacf(precip_train2, lag.max=72)
```

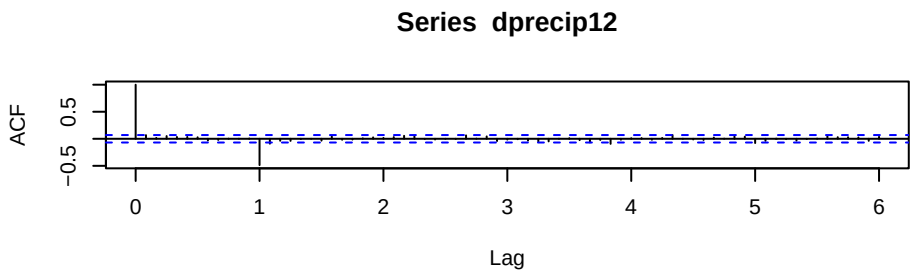


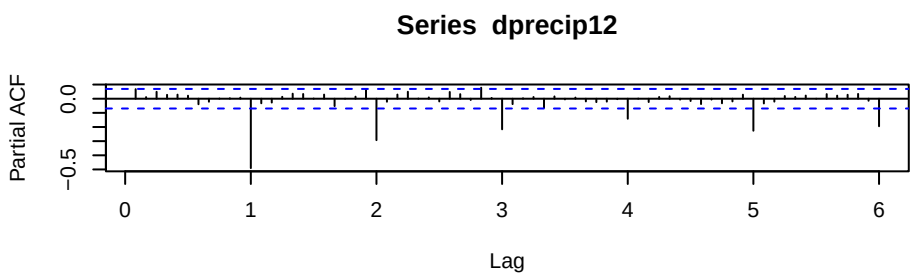

```
precip_acfv = c(precip_acf$acf)
lags = 1+seq(6,72,6)  # every 6th serial correlation, first is 1 for lag 0

# acf values for total precipitation at lags of 6 months apart
print(round(precip_acfv[lags],3))
#>  [1] -0.440  0.553 -0.456  0.535 -0.440  0.501 -0.428  0.489 -0.430  0.463
#> [11] -0.424  0.511

# Decrease in absolute value is very slow; Box-Jenkins ARIMA suggest seasonal differencing

acf12 = precip_acfv[13]  # lag12 serial correlation
print(round(acf12^(1:6),3))
#> [1] 0.553 0.306 0.169 0.094 0.052 0.029

# Comparing the two sequences suggest that a non-stationary model should be used

# Winters additive seasonal exponential smoothing
aseas_precip = HoltWinters(precip_train, seasonal="additive")

print(aseas_precip)
#> Holt-Winters exponential smoothing with trend and additive seasonal component.
#>
#> Call:
```

```
#> HoltWinters(x = precip_train, seasonal = "additive")
#>
#> Smoothing parameters:
#>  alpha: 0.0593614
#>  beta : 0.01002445
#>  gamma: 0.11516
#>
#> Coefficients:
#>             [,1]
#> a   113.64172224
#> b     0.00636881
#> s1   57.51643851
#> s2  -23.84900264
#> s3   -0.54188835
#> s4  -30.26468843
#> s5  -47.85416218
#> s6  -64.82984693
#> s7  -74.65446044
#> s8  -75.99370982
#> s9  -55.65963740
#> s10  17.90764375
#> s11  55.85163910
#> s12  57.95612123


# Diagnostics for differencing at lag 12
dprecip12 = diff(precip_train, lag=12)

par(mfrow=c(3,1))
plot(dprecip12)
print(window(dprecip12,start=c(1939,1),end=c(1948,12)))
#>         Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct
#> 1939   97.9    7.8  -19.2  -36.9    0.2   31.3   33.0    3.8   14.8    2.9
#> 1940 -104.6   51.1   61.8   42.3    8.9  -40.7   -5.9   15.4    7.1   74.7
#> 1941   29.7  -14.7  -51.0  -10.2   40.6   37.6  -29.0   17.5   67.0  -40.5
#> 1942  -68.2  -77.1   -4.0   -2.1  -62.3   40.4   38.3  -39.1 -114.3  -37.3
#> 1943    3.3   40.9    1.7   18.9   27.5  -67.8  -24.2   45.7   30.6   33.8
#> 1944   72.2   -1.7  -28.0  -12.0  -34.8    2.5  -13.9  -35.7   56.4  -10.6
#> 1945   13.4   32.8   81.1    5.4   -7.0   -5.3   12.0   11.5  -21.2   15.8
#> 1946   25.6   36.5   28.4   60.5  -17.2   83.5   22.0  -15.1  -40.7  -27.5
#> 1947  -25.7  -18.5  -32.5  -37.3   28.4  -52.0   -4.1   -7.0   18.1   66.0
#> 1948  -71.1   45.1  -77.0  -35.4   84.0  -12.9   10.9   67.9    2.1  -98.3
#>         Nov    Dec
#> 1939   96.6   -7.2
#> 1940 -101.2  -15.4
#> 1941   42.4  -56.3
#> 1942   -1.1   30.4
#> 1943  -61.5  -65.4
#> 1944   95.0  -48.5
#> 1945   33.8   64.9
#> 1946  -78.7    5.3
#> 1947  -17.5   96.4
#> 1948  152.4  -70.7
```

```r
# Does this series appear stationary?

dprecip12_acf = acf(dprecip12, lag.max=72)
dprecip12_pacf = pacf(dprecip12, lag.max=72)
```

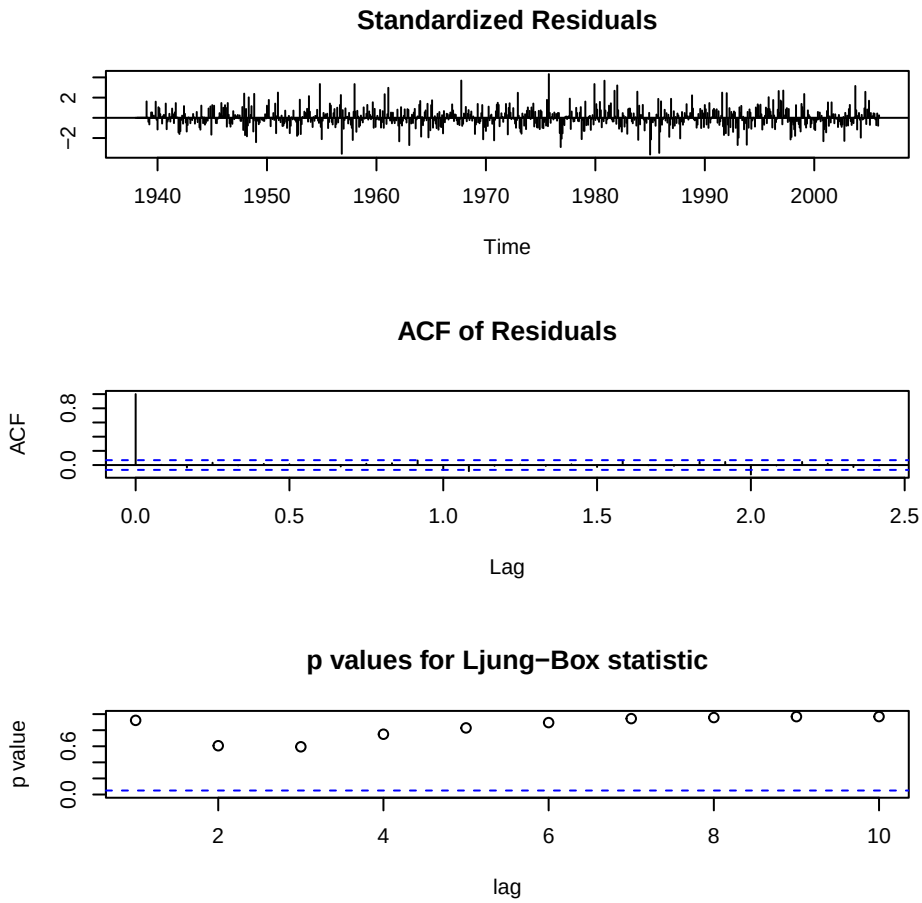




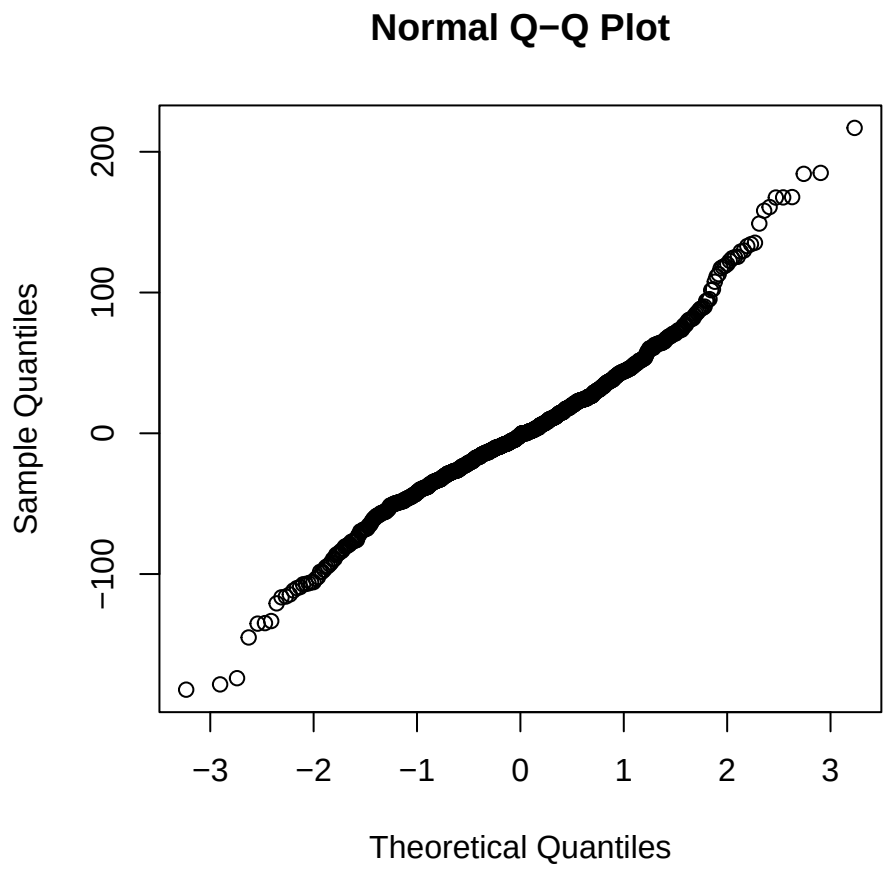



```r
dprecip12_acfv = c(dprecip12_acf$acf)
dprecip12_pacfv = c(dprecip12_pacf$acf)

# acf values for seasonal difference of precipitation at lags of 6 months apart
print(round(dprecip12_acfv[lags],3))
#>  [1]  0.028 -0.487 -0.035  0.016  0.000 -0.015  0.011  0.010 -0.013 -0.079
#> [11]  0.003  0.036

print(round(dprecip12_pacfv[lags-1],3))
#>  [1]  0.021 -0.491  0.001 -0.293 -0.019 -0.216 -0.005 -0.142 -0.016 -0.226
#> [11]  0.000 -0.195

# Compare fits of stationary and nonstationary models
auto_precip_stationary = auto.arima(precip_train, seasonal=T,stationary=T)

auto_precip_nonstationary = auto.arima(precip_train, seasonal=T,stationary=F)
```

```r
print(auto_precip_stationary)
#> Series: precip_train
#> ARIMA(1,0,0)(2,0,0)[12] with non-zero mean
#>
#> Coefficients:
#>          ar1    sar1    sar2     mean
#>       0.1414  0.3450  0.3188  93.4710
#> s.e.  0.0384  0.0338  0.0345   5.7775
#>
#> sigma^2 = 2477:  log likelihood = -4347.34
#> AIC=8704.69   AICc=8704.76   BIC=8728.21

print(auto_precip_nonstationary)
#> Series: precip_train
#> ARIMA(1,0,1)(2,1,0)[12]
#>
#> Coefficients:
#>          ar1      ma1     sar1     sar2
#>       0.8296  -0.7731  -0.6362  -0.2979
#> s.e.  0.0990   0.1114   0.0341   0.0340
#>
#> sigma^2 = 2518:  log likelihood = -4289.7
#> AIC=8589.4   AICc=8589.47   BIC=8612.84

# Note that the fitted model with seasonal differencing has smaller AIC

# tsdiag function is used to check if diagnostics suggest that estimated innovations
# behave close to white noise
tsdiag(auto_precip_nonstationary)
```



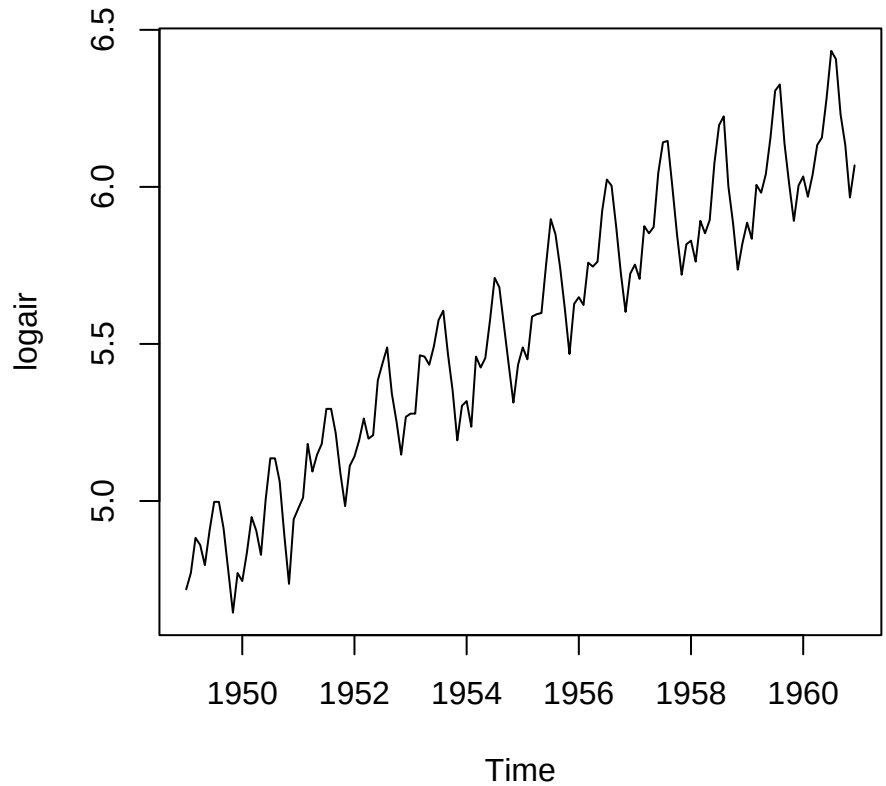

```
par(mfrow=c(1,1))
qqnorm(auto_precip_nonstationary$residuals)
```




```
# Estimated innovations have weak serial correlations
round(c(acf(auto_precip_nonstationary$residuals,plot=F,lag.max=12)$acf),3)
#>  [1]  1.000  0.003 -0.035  0.033 -0.005  0.017  0.012 -0.002 -0.020  0.019
#> [11]  0.026  0.064 -0.070
```

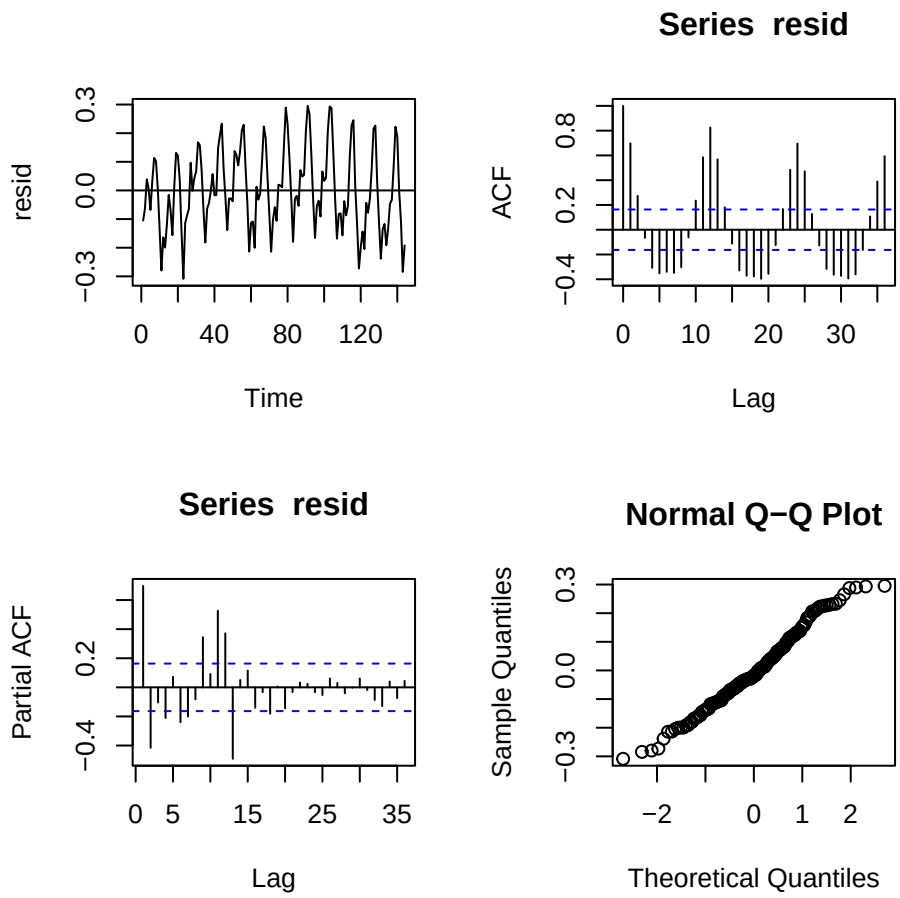
data(AirPassengers) with logarithm because of increasing variance over time

```
# Data set of monthly totals of international airline passengers, 1949 to 1960,
# from book by Box & Jenkins
data(AirPassengers)
logair = log(AirPassengers)
plot(logair)
```




```r
summary(AirPassengers) # in thousands
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>   104.0   180.0   265.5   280.3   360.5   622.0

ntotal = length(logair)
print(ntotal)
#> [1] 144
tt = 1:ntotal

# Regress out the linear trend and model the deviations from the trend
reg = lm(logair ~ tt)
summary(reg)
#>
#> Call:
#> lm(formula = logair ~ tt)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -0.30858 -0.10388 -0.01796  0.09738  0.29538
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) 4.8136683  0.0232940  206.65   <2e-16 ***
```

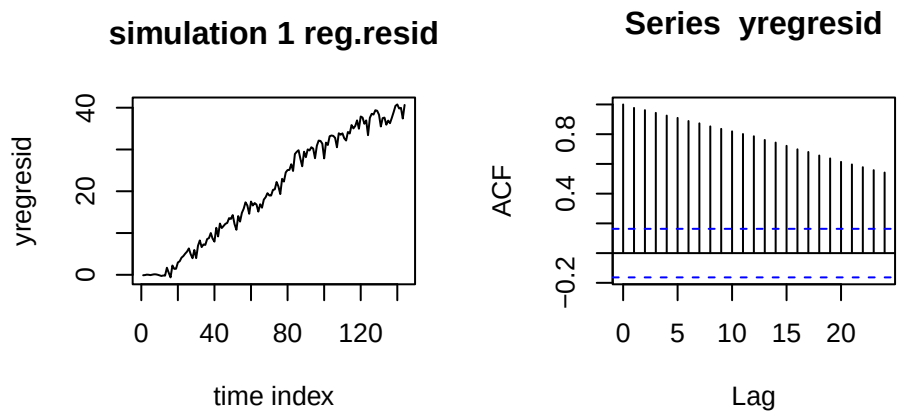
```
#> tt          0.0100484  0.0002787   36.05   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.139 on 142 degrees of freedom
#> Multiple R-squared:  0.9015, Adjusted R-squared:  0.9008
#> F-statistic:  1300 on 1 and 142 DF,  p-value: < 2.2e-16

resid = reg$residuals
par(mfrow=c(2,2))
plot.ts(resid); abline(h=0)
resid_acf = acf(resid, lag.max=36)
pacf(resid, lag.max=36)
qqnorm(resid)
```

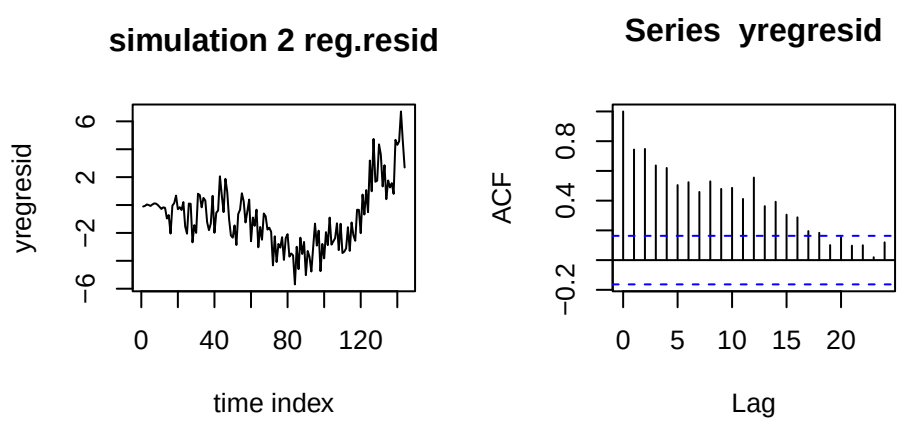



```
# Winters additive seasonal exponential smoothing
resid_ts = ts(resid,start=c(1,1),frequency=12)
aseas_resid = HoltWinters(resid_ts, seasonal="additive")

print(aseas_resid)
#> Holt-Winters exponential smoothing with trend and additive seasonal component.
#>
```

```
#> Call:
#> HoltWinters(x = resid_ts, seasonal = "additive")
#>
#> Smoothing parameters:
#>  alpha: 0.3285654
#>  beta : 0.00246745
#>  gamma: 0.7730104
#>
#> Coefficients:
#>             [,1]
#> a   -0.087852377
#> b   -0.001555519
#> s1  -0.073713240
#> s2  -0.140301651
#> s3  -0.032431170
#> s4   0.014709109
#> s5   0.034351575
#> s6   0.158113189
#> s7   0.296036761
#> s8   0.277737860
#> s9   0.088455860
#> s10 -0.034139390
#> s11 -0.198382649
#> s12 -0.103336567

# auto.arima applied to the regression residuals
# The beta parameter in Winters additive smoothing is small but arima
# has differencing at lag 1 and lag 12

resid_auto = auto.arima(resid_ts,seasonal=T, stationary=F)
print(resid_auto)
#> Series: resid_ts
#> ARIMA(0,1,1)(0,1,1)[12]
#>
#> Coefficients:
#>           ma1     sma1
#>       -0.4018  -0.5569
#> s.e.   0.0896   0.0731
#>
#> sigma^2 = 0.001369:  log likelihood = 244.7
#> AIC=-483.39   AICc=-483.2   BIC=-474.77

# Simulation for this model

par(mfrow=c(2,2))
s = sqrt(resid_auto$sigma2)
coe = resid_auto$coef
ma_coe = c(coe[1],rep(0,10),coe[2],coe[1]*coe[2])

ystart = resid[1:13]
ntotal = 144 # 12 years

set.seed(123)
```

```r
# Compare 3 simulations
for(isim in 1:2)
{ zma = rep(0,ntotal)
  # Generate  ARMA(0,1)(0,1)_{12} sequence
  zma = arima.sim(n=ntotal,list(ma=ma_coe,sd=s))
  # Seasonal ARIMA with lag 1 and lag12 differencing
  yregresid = rep(0,ntotal)
  yregresid[1:13] = ystart
  for(i in 14:ntotal)
  { yregresid[i] = yregresid[i-1] + yregresid[i-12] - yregresid[i-13] + zma[i] }
  plot.ts(yregresid, xlab="time index", main=paste("simulation",isim,"reg.resid"))
  acf(yregresid,lag.max=24)
}
```

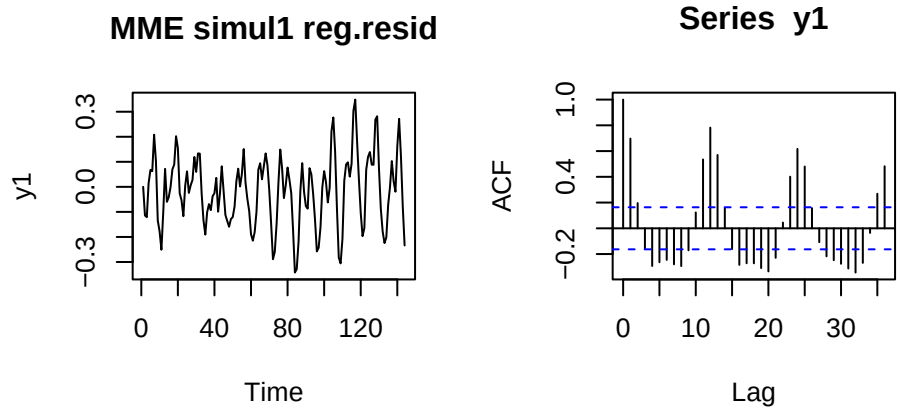


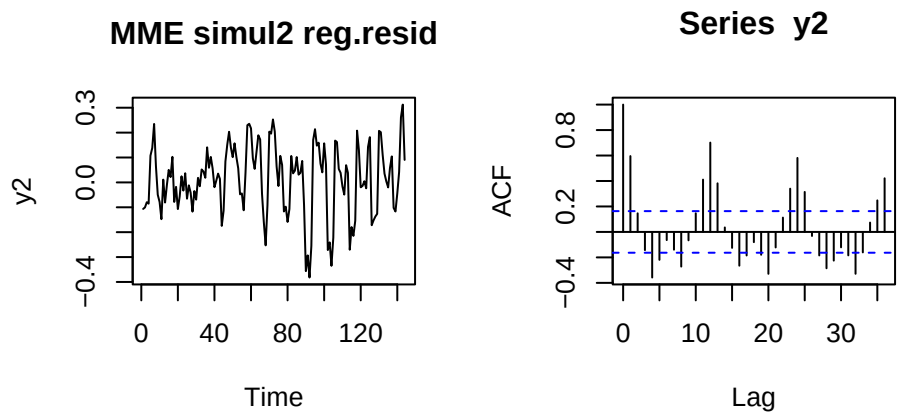



```r
# Alternative time series model that generates simulated series
# closer in appearance to the original.

#resid_ar36 = arima(resid,order=c(36,0,0),include.mean=F, method="CSS")
# non-stationary phi polynomial from the regression on past 36 observations


# MME based on acf2AR is used, because CSS and ML can estimate AR(36)
toep = toeplitz(c(resid_acf$acf))
```

```r
# eigenvalues should all be positive if Toeplitz matrix is positive
tem = eigen(toep)$values
print(round(tem,3))
#>  [1] 10.690 10.242  3.165  3.053  1.810  1.363  0.762  0.714  0.624  0.581
#> [11]  0.439  0.428  0.419  0.296  0.283  0.272  0.263  0.259  0.255  0.249
#> [21]  0.158  0.104  0.081  0.075  0.060  0.057  0.043  0.036  0.035  0.031
#> [31]  0.028  0.023  0.023  0.023  0.021  0.017  0.017

armat = acf2AR(c(resid_acf$acf))

mme36 = armat[36,1:36]
round(mme36,4)
#>       1       2       3       4       5       6       7       8       9      10
#>  0.8497 -0.2153 -0.0586 -0.0873  0.1240  0.0040 -0.0283 -0.0944  0.2891 -0.2060
#>      11      12      13      14      15      16      17      18      19      20
#>  0.0654  0.6329 -0.4053 -0.1168  0.1292 -0.0610  0.0927 -0.2374  0.0795 -0.0679
#>      21      22      23      24      25      26      27      28      29      30
#>  0.0009 -0.0458  0.1109  0.0357 -0.1490  0.0636  0.0578 -0.0163 -0.0766  0.0598
#>      31      32      33      34      35      36
#>  0.0202  0.0296 -0.1744  0.1118 -0.1113  0.0442

# Checks if the AR polynomial has roots all outside the complex unit circle
print(abs(polyroot(c(1,-mme36))))
#>  [1] 1.010864 1.085759 1.014783 1.007316 1.072190 1.014783 1.012891 1.088422
#>  [9] 1.035217 1.088301 1.030654 1.012891 1.052127 1.010864 1.007316 1.010417
#> [17] 1.075802 1.059909 1.063424 1.087306 1.010417 1.063424 1.059909 1.075802
#> [25] 1.087306 1.088301 1.035217 1.079565 1.030654 1.079565 1.085759 1.072190
#> [33] 1.088422 1.445769 1.445769 2.099047

# The AR(36) model is stationary but close to the boundary of non-stationary.

# Simulation of deviation from linear with AR(36) stationary model
sigmae = sd(resid)
set.seed(12345)
par(mfrow=c(2,2))

y1 = arima.sim(n=ntotal,list(ar=mme36,sd=1))
y1 = y1*sigmae/sd(y1)
plot.ts(y1, main="MME simul1 reg.resid")
acf(y1,lag.max=36)
y1_mat = matrix(y1,byrow=T,ncol=12)
round(10*y1_mat,2)
#>        [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7] [,8]  [,9] [,10] [,11] [,12]
#>  [1,]  0.00 -1.14 -1.21  0.15  0.68  0.63  2.08 1.02 -1.34 -1.78 -2.51 -1.01
#>  [2,]  0.72 -0.60 -0.42  0.00  0.70  0.89  2.02 1.55 -0.26 -0.53 -1.17  0.04
#>  [3,]  0.62 -0.24  0.06  0.28  1.19  0.60  1.33 1.33 -0.27 -1.33 -1.91 -1.02
#>  [4,] -0.69 -0.93 -0.37 -0.25  0.36 -0.99 -0.28 0.82 -0.17 -1.12 -1.36 -1.59
#>  [5,] -1.29 -1.21 -0.79  0.23  0.73  0.01  0.51 1.51  0.16 -0.48 -0.97 -1.89
#>  [6,] -2.15 -1.81 -0.99  0.68  0.94  0.30  0.82 1.34  0.86 -0.17 -1.46 -2.89
#>  [7,] -2.60 -1.47  0.18  1.49  0.73 -0.45  0.07 0.77  0.29 -0.25 -1.73 -3.42
#>  [8,] -3.29 -2.19 -0.15  0.95 -0.20 -0.77 -0.87 0.75  0.50 -0.29 -1.31 -2.58
#>  [9,] -2.43 -1.60 -0.02  0.63  0.19 -0.62 -0.04 2.21  2.77  1.40 -0.88 -2.83
#> [10,] -3.06 -2.08  0.24  0.90  0.98  0.41  0.92 3.03  3.48  1.83  0.30 -1.03
```

```
#> [11,] -1.97 -1.64  0.69  1.19  1.39  0.89  0.88 2.68  2.82  0.70 -0.88 -1.78
#> [12,] -2.24 -2.02 -0.72 -0.01  1.03  0.22 -0.19 1.62  2.72  1.21 -0.79 -2.34

# Compare with the regression residuals: what is main difference?
regresid_mat = matrix(resid,byrow=T,ncol=12)
round(10*regresid_mat,2)
#>        [,1]  [,2]  [,3]  [,4]  [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12]
#>  [1,] -1.05 -0.63  0.39  0.06 -0.68 0.31 1.13 1.03 0.09 -1.35 -2.80 -1.64
#>  [2,] -1.99 -1.18 -0.16 -0.69 -1.56 0.09 1.31 1.21 0.38 -1.44 -3.09 -1.13
#>  [3,] -0.88 -0.64  0.97 -0.01  0.42 0.67 1.68 1.58 0.70 -0.68 -1.82 -0.63
#>  [4,] -0.44 -0.03  0.57 -0.17 -0.16 1.49 1.92 2.33 0.76 -0.24 -1.38 -0.28
#>  [5,] -0.28 -0.38  1.38  1.23  0.87 1.37 2.10 2.29 0.82 -0.45 -2.14 -1.13
#>  [6,] -1.08 -2.00  0.13 -0.32 -0.11 0.99 2.24 1.83 0.50 -0.83 -2.14 -1.03
#>  [7,] -0.58 -1.06  0.20  0.17  0.11 1.55 2.90 2.32 1.15 -0.25 -1.80 -0.30
#>  [8,] -0.19 -0.54  0.71  0.48  0.54 2.06 2.95 2.66 1.24 -0.35 -1.66 -0.55
#>  [9,] -0.36 -0.91  0.66  0.34  0.44 2.06 2.93 2.88 1.33 -0.29 -1.69 -0.82
#> [10,] -0.80 -1.57 -0.37 -0.87 -0.55 1.16 2.27 2.45 0.12 -1.16 -2.73 -1.99
#> [11,] -1.43 -2.05 -0.43 -0.78 -0.29 0.77 2.16 2.26 0.28 -1.11 -2.38 -1.36
#> [12,] -1.17 -1.91 -1.32 -0.47 -0.33 0.82 2.23 1.86 0.00 -1.07 -2.84 -1.92

y2 = arima.sim(n=ntotal,list(ar=mme36,sd=1))
y2 = y2*sigmae/sd(y2)
plot.ts(y2, main="MME simul2 reg.resid")
acf(y2,lag.max=36)
```






```
y2_mat = matrix(y2,byrow=T,ncol=12)
round(10*y2_mat,2)
#>        [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10] [,11] [,12]
#>  [1,] -1.07 -1.00 -0.80 -0.86  1.07  1.39  2.35  0.65 -0.50 -0.79 -1.48  0.11
#>  [2,] -0.81 -0.10  0.50  0.21  1.03 -0.79 -0.18 -1.07 -0.56  0.24 -0.33  0.37
#>  [3,] -0.65 -0.12 -0.42 -1.17 -0.35 -0.69  0.18 -0.15  0.52  0.44  0.19  1.41
#>  [4,]  0.59  1.02  0.56 -0.19  0.07  0.35  0.14 -1.75 -1.14  0.84  1.51  2.03
#>  [5,]  1.39  1.01  1.57  1.02  0.57 -0.48 -0.44 -1.12  0.39  2.30  2.35  2.17
#>  [6,]  0.97  0.54  1.23  1.89  1.73 -0.45 -1.59 -2.53 -0.91  2.04  1.96  2.53
#>  [7,]  2.04  0.64 -0.15  1.07  0.70 -1.17 -0.98 -1.59 -1.07  1.06  0.36  0.47
#>  [8,]  1.02  0.31  0.38  0.86 -0.91 -3.56 -2.93 -3.82 -2.55  1.70  2.14  1.48
#>  [9,]  1.60  0.75  0.40  1.57  0.72 -2.72 -2.41 -3.35 -2.22  1.68  1.62  0.54
#> [10,]  0.38 -0.16 -0.49  0.69  0.38 -2.71 -1.80 -2.15 -1.54  2.08  1.30 -0.20
#> [11,] -0.14  0.05 -0.24  1.42  1.82 -1.72 -1.53 -1.37 -1.26  2.07  2.02  1.11
#> [12,]  0.35  0.04 -0.20  0.58  1.04 -1.02 -1.17 -0.43  0.42  2.61  3.13  0.90

# Compare with the regression residuals: what is main difference?

# ARIMA with twice differencing should be good enough for short term forecasting
# but it does not generating data series that look like the actual data
```